1. Renormalization in ϕ^3-theory

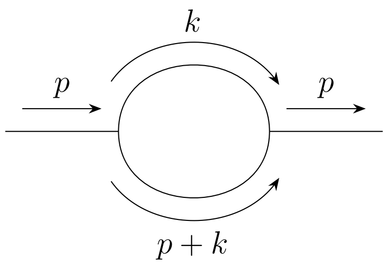
One loop of self energy

Simply speaking, renormalization is to solve the problem of divergent when we try to solve the loop graphs by subtracting the divergent part in the expression. In the following, we will use ϕ^3-theory as an example to discuss the renormalization.

Firstly, the Lagrangian of ϕ^3-theory is:

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3. \tag{1}$$

As the simplest case, the 1-loop correlation for the propagator is (notice that in this graph ,there are **all inner lines**, i.e., propagators):



where the momentum k is **off-shelled**, which means k doesn't follow the relation $k^2 + E_k^2 = m^2$, so technically k could be very large. But we don't need to worry about it, because k only appears in the loop, which means that it must follows the 4-momentum conservation. Thus we need to sum over all possible k values, which finally will contribute a integral measurement when we calculate the loop part. Thus, generally, we can write the whole graph as

$$\frac{\mathrm{i}}{p^2 - m^2 + \mathrm{i}\epsilon}\left[-\mathrm{i}\Sigma_1(p^2)\right]\frac{\mathrm{i}}{p^2 - m^2 + \mathrm{i}\epsilon}, \tag{2}$$

where $-\mathrm{i}\Sigma_1(p^2)$ represent the loop part, i.e.,

$$-\mathrm{i}\Sigma_1(p^2) = \frac{g^2}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(p+k)^2 - m^2 + \mathrm{i}\epsilon}, \tag{3}$$

where we will note the integral measurement $\frac{\mathrm{d}^4k}{(2\pi)^4} \to \mathrm{d}^4k$ for simple. Next, we will focus on this integral as an example to talk about how to solve this type of integral.

1.1 Wick rotation

The integral in (3) over the loop energy can be also written as

$$I(p, \vec{k}) = \int_{-\infty}^{+\infty} \mathrm{d}k^0 \frac{\mathrm{i}}{(k^0)^2 - E_k^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(p^0 + k^0)^2 - E_{p+k}^2 + \mathrm{i}\epsilon}, \tag{4}$$

where

$$E_k = \sqrt{\vec{k}^{\,2} + m^2}, \qquad E_{p+k} = \sqrt{(\vec{p} + \vec{k})^2 + m^2}. \tag{5}$$

The variable k^0 is now regarded as a complex variable. The four poles are

$$k^0 = +E_k - \mathrm{i}\epsilon, \qquad k^0 = -E_k + \mathrm{i}\epsilon, \tag{6}$$
$$k^0 = -p^0 + E_{p+k} - \mathrm{i}\epsilon, \qquad k^0 = -p^0 - E_{p+k} + \mathrm{i}\epsilon. \tag{7}$$

The small $\mathrm{i}\epsilon$ is important: it tells us which poles sit slightly above or slightly below the real axis. Wick rotation means changing the integration path in the complex k^0 plane. Originally the integral is along the real axis,

$$C_{\mathrm{M}}: \quad k^0 \in (-\infty, +\infty).$$

In Euclidean variables we use

$$k^0 = \mathrm{i}\omega, \qquad \mathrm{d}k^0 = \mathrm{i}\,\mathrm{d}\omega,$$

so the new path is the imaginary axis,

$$C_{\mathrm{E}}: \quad k^0 = \mathrm{i}\omega, \quad \omega \in (-\infty, +\infty).$$

In the figures below, the blue path is the real-axis Minkowski contour C_{M}, the green path is the imaginary-axis Euclidean contour C_{E}, and the crosses are poles.

There is one orientation point which is very easy to mix up. The Euclidean contour itself is oriented upward, because

$$\omega : -\infty \to +\infty \quad \Longrightarrow \quad k^0 = \mathrm{i}\omega : -\mathrm{i}\infty \to +\mathrm{i}\infty.$$

But when we prove the Wick rotation by closing the contour, the closed contour is

$$C_{\mathrm{closed}} = C_{\mathrm{M}} + A_+ - C_{\mathrm{E}} + A_-,$$

where A_+ runs from $+R$ to $+\mathrm{i}R$ in the first quadrant, while A_- runs from $-\mathrm{i}R$ back to $-R$ in the third quadrant. Thus the imaginary-axis part of the closed contour is the downward path $-C_{\mathrm{E}}$. After moving it to the other side of the equation, the final Euclidean integral is again along C_{E}, upward. The orange dashed arrows in the figures show these auxiliary closing arcs, not the direction of the final integral along C_{E}.

The phrase "change the contour" means the following simple thing: we slide the integration line from one path to another in the complex k^0 plane. If the sliding line does not hit any pole, the integral is unchanged, apart from the Jacobian $\mathrm{d}k^0 = \mathrm{i}\,\mathrm{d}\omega$. If the line hits a pole, then we must bend the path around that pole, and the difference is a residue term. Symbolically,

$$\int_{C_{\mathrm{old}}} f(k^0)\,\mathrm{d}k^0 - \int_{C_{\mathrm{new}}} f(k^0)\,\mathrm{d}k^0 = 2\pi\mathrm{i} \sum_{\text{poles swept by the contour}} \operatorname{Res} f, \tag{8}$$

up to the sign determined by the orientation of the contour.

Case 1: p^0 is imaginary.

Take, for example, $p^0 = -\mathrm{i}P$ with $P > 0$. Then $-p^0 = +\mathrm{i}P$, so the two poles from the second propagator are shifted upward:

$$k^0 = \mathrm{i}P + E_{p+k} - \mathrm{i}\epsilon, \qquad k^0 = \mathrm{i}P - E_{p+k} + \mathrm{i}\epsilon.$$

This is the Euclidean starting point. The current integration path is C_{E}, not C_{M}.

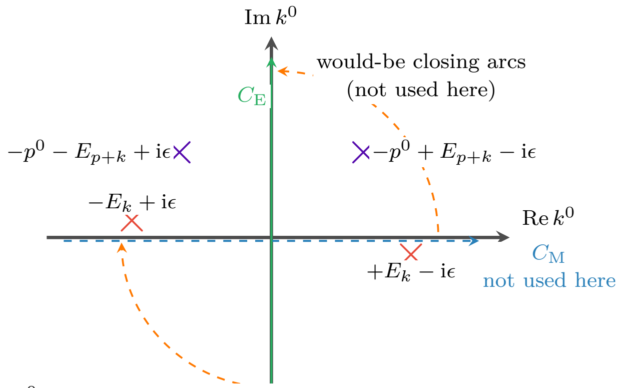


Fig. 1: p^0 imaginary. The first-quadrant pole is harmless, because the contour currently being used is only the green line C_E.

The point which is easy to miss is this: the pole in the first quadrant does not automatically give a residue. A residue appears only when the contour is actually moved through that pole, or when the pole moves through the contour. In this first figure we are not rotating C_E into C_M while keeping p^0 imaginary. We are simply defining the Euclidean integral on the green line.

Case 2: p^0 is real and $|p^0| < m$.

Now keep the green contour C_E fixed, and analytically continue p^0 from imaginary value back to a real value. The poles move because their positions contain p^0. If $|p^0| < m$, then

$$E_{p+k} \geq m > |p^0|,$$

so

$$-p^0 + E_{p+k} > 0, \qquad -p^0 - E_{p+k} < 0.$$

Thus the second pair of poles ends up in the same pattern as the first pair: one pole is slightly below the real axis on the right, and one pole is slightly above the real axis on the left.

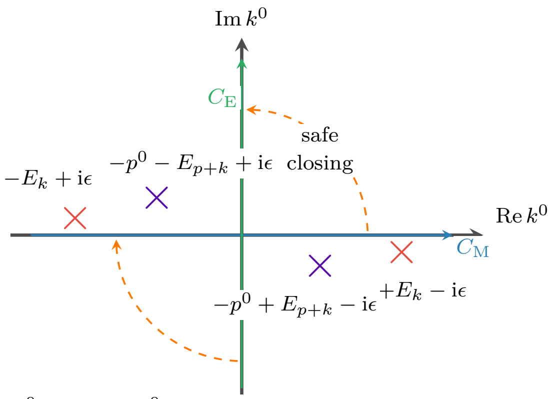


Fig. 2: p^0 real and $|p^0| < m$. No pole lies in the region swept when the blue contour is rotated into the green contour.

Therefore, for $|p^0| < m$, the real-axis integral and the imaginary-axis integral are related by an ordinary Wick rotation. On the green contour,

$$k^0 = i\omega, \qquad k^2 = (k^0)^2 - \vec{k}^2 = -(\omega^2 + \vec{k}^2) < 0,$$

and similarly $(p+k)^2 < 0$ in the large Euclidean region. This is why power counting becomes Euclidean.

What does “a pole crosses a contour” mean?

A pole is not just a fixed dot: its position depends on the external energy p^0. For example, if $p^0 = -P$ with $P > 0$, then one pole is

$$k^0 = -p^0 - E_{p+k} + \mathrm{i}\epsilon = P - E_{p+k} + \mathrm{i}\epsilon. \tag{9}$$

As P increases, this pole moves horizontally. If $P < E_{p+k}$, it is on the left side of the imaginary axis. If $P = E_{p+k}$, it sits on the imaginary-axis contour. If $P > E_{p+k}$, it has moved to the right side of the imaginary axis. This is what we mean by saying that the pole has crossed C_{E}.

In formula language, crossing the imaginary axis means

$$\operatorname{Re} k^0_{\text{pole}} \quad \text{changes sign.}$$

Crossing the real axis would similarly mean

$$\operatorname{Im} k^0_{\text{pole}} \quad \text{changes sign.}$$

But only crossing the contour that we are actually using produces a residue.

Case 3: p^0 is real and $|p^0| > m$.

Take the negative-energy example $p^0 = -P$ with $P > m$, which is the case drawn in the book. The pole

$$k^0 = P - E_{p+k} + \mathrm{i}\epsilon$$

crosses the green contour whenever

$$P > E_{p+k} \quad \Longleftrightarrow \quad (p^0)^2 > m^2 + (\vec{p} + \vec{k})^2.$$

When this happens, the contour cannot pass straight through the pole. It must detour around it, and this detour gives an extra residue, usually called the pole term.

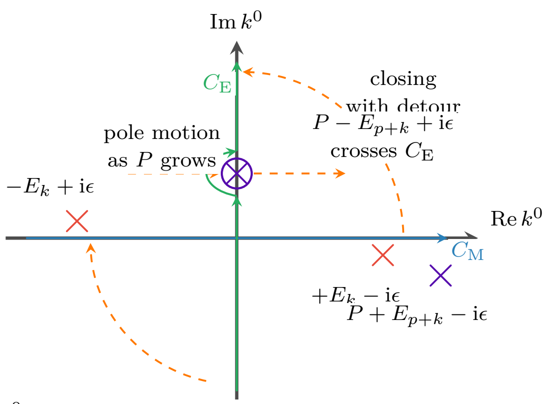


Fig. 3: $p^0 = -P$ and $P > m$. A pole crosses the green contour, so the Wick-rotated answer contains an extra residue term.

However, this pole term can occur only when

$$(p^0)^2 > m^2 + (\vec{p} + \vec{k})^2.$$

For very large $|\vec{k}|$, the right-hand side is very large, so the inequality cannot hold. Therefore the pole term comes only from a finite region of $\vec{k}$, not from the UV region $|\vec{k}| \to \infty$. The UV divergence is still captured by the Euclidean large-momentum integral along C_{E}.

So the divergence in Eq.(3) is called the UV divergence, which can be seen by firstly taking the polar coordinate: $\mathrm{d}^4k_E = k_E^3 \mathrm{d}k_E \mathrm{d}\Omega$, thus in large k

$$\Sigma_1 \sim \int \mathrm{d}^d k \frac{1}{k^4} \sim \int \mathrm{d}k \mathrm{d}\Omega \frac{1}{k} \sim \int \frac{\mathrm{d}k}{k} \sim \ln \Lambda \to \infty. \tag{10}$$

1.2 On-shell renormalization

Renormalization begins by separating a loop integral into two parts: a finite, momentum-dependent part and a divergent part that is local. For the one-loop self-energy in ϕ^3 theory,

$$\Sigma_1(p^2, m^2) = \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4k \, \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{11}$$

The integral is logarithmically divergent in the ultraviolet. A convenient way to display the divergent piece is to add and subtract a simple reference integral. We introduce an arbitrary finite mass scale μ and write

$$\begin{aligned}\Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4} \Bigg[\int \mathrm{d}^4k \Bigg\{ &\frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \Bigg\} \\ &+ \int \mathrm{d}^4k \, \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \Bigg].\end{aligned} \tag{12}$$

This is just adding zero, so no physics has been changed. The reason for this particular subtraction is that the first integral now falls faster at large loop momentum:

$$\frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} = O\left(\frac{1}{k^6}\right). \tag{13}$$

In four dimensions, $\mathrm{d}^4k \sim k^3 \mathrm{d}k$, and therefore

$$\int^{\infty} \mathrm{d}^4k \, O\left(\frac{1}{k^6}\right) \sim \int^{\infty} \mathrm{d}k \, \frac{1}{k^3}, \tag{14}$$

which is ultraviolet finite. Thus the logarithmic divergence has been isolated in the second integral. Since that second integral has no dependence on the external momentum p, its divergent part can be canceled by a local mass counterterm.

To see the divergence explicitly, regulate the theory on a spacetime lattice with lattice spacing a. The lattice spacing acts as a UV cutoff of order $1/a$. If $S_F(k; m, a)$ denotes the regulated free propagator, then

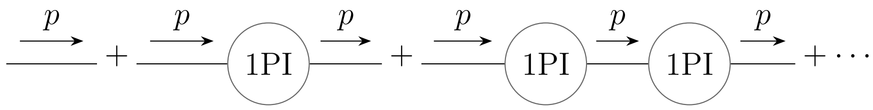
$$\begin{aligned}\Sigma_1 &= \frac{\mathrm{i}g^2}{32\pi^4} \left[\int \mathrm{d}^4k \, \{ S_F(k; m, a) S_F(p+k; m, a) - S_F^2(k; \mu, a) \} + \int \mathrm{d}^4k \, S_F^2(k; \mu, a) \right] \\ &\equiv \Sigma_{1,\mathrm{fin}}(p, m, \mu, a) + \Sigma_{1,\mathrm{div}}(\mu, a).\end{aligned} \tag{15}$$

The divergent reference part is

$$\begin{aligned}\Sigma_{1,\mathrm{div}}(\mu, a) &= \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4k \, S_F^2(k; \mu, a) \\ &= -\frac{\mathrm{i}g^2}{32\pi^4} \int_{\omega^2 + \vec{k}^2 < 1/a^2} \mathrm{d}\omega \, \mathrm{d}^3\vec{k} \, \frac{1}{(\omega^2 + \vec{k}^2 + \mu^2)^2} \\ &= -\frac{\mathrm{i}g^2}{16\pi^2} \int_0^{1/a} \mathrm{d}K \, \frac{K^3}{(K^2 + \mu^2)^2} \\ &= -\frac{\mathrm{i}g^2}{16\pi^2} \ln \frac{1}{a} + \text{finite terms}, \qquad a \to 0.\end{aligned} \tag{16}$$

This agrees with the power-counting estimate in Eq. (10): the one-loop two-point graph in four-dimensional ϕ^3 theory is logarithmically divergent.

The same divergence can be understood from the full propagator. Let $-\mathrm{i}\Sigma(p^2)$ denote the sum of all one-particle-irreducible two-point insertions. Repeated insertions give

$G_2(p^2) =$ 

$$\begin{aligned} &= \frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \cdots \\ &= \frac{\mathrm{i}}{p^2-m^2}\left[1 + \frac{\Sigma(p^2)}{p^2-m^2} + \left(\frac{\Sigma(p^2)}{p^2-m^2}\right)^2 + \cdots\right] \\ &= \frac{\mathrm{i}}{p^2-m^2-\Sigma(p^2)}. \end{aligned} \tag{17}$$

The physical mass is defined by the pole of the full propagator:

$$m_{\mathrm{ph}}^2 - m^2 - \Sigma(m_{\mathrm{ph}}^2) = 0, \qquad m_{\mathrm{ph}}^2 = m^2 + \Sigma(m_{\mathrm{ph}}^2). \tag{18}$$

This equation fixes only the pole position. Away from the pole, the full propagator is not simply $\mathrm{i}/(p^2 - m_{\mathrm{ph}}^2)$; the remaining momentum dependence of $\Sigma(p^2)$ still contributes.

In the on-shell scheme the renormalized mass parameter is chosen to be the physical pole mass. Thus the loop integral is written with m_{ph}:

$$\Sigma_1(p^2) = \frac{\mathrm{i}g^2}{32\pi^4}\int \mathrm{d}^4k\, \frac{1}{(k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)((p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)}. \tag{19}$$

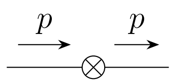
The bare mass is then split as

$$m_0^2 = m_{\mathrm{ph}}^2 + \delta m^2. \tag{20}$$

In the Lagrangian this means

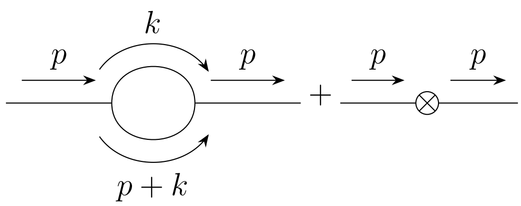
$$-\frac{1}{2}m_0^2\phi^2 = -\frac{1}{2}m_{\mathrm{ph}}^2\phi^2 - \frac{1}{2}\delta m^2\phi^2. \tag{21}$$

The counterterm vertex is therefore



$$= -\mathrm{i}\,\delta m^2. \tag{22}$$

The renormalized one-loop self-energy is the sum of the ordinary bubble and the counterterm graph:

$-\mathrm{i}\Sigma_{1R}(p^2) =$ 

$$= -\mathrm{i}\Sigma_1(p^2) - \mathrm{i}\delta m^2. \tag{23}$$

Equivalently,

$$\Sigma_{1R}(p^2) = \Sigma_1(p^2) + \delta m^2 = \Sigma_{1,\mathrm{fin}}(p^2) + \left(\Sigma_{1,\mathrm{div}} + \delta m^2\right). \tag{24}$$

Choosing

$$\delta m^2 = -\Sigma_{1,\mathrm{div}} \tag{25}$$

cancels the divergent part. In the on-shell scheme, the remaining finite part is fixed by the condition that the pole sits at $p^2 = m_{\mathrm{ph}}^2$.

A useful way to compute the finite answer is to differentiate with respect to p^2. The counterterm and the subtracted reference integral do not depend on p, so the derivative acts only on the original loop integral:

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= \frac{p^\mu}{2p^2}\frac{\partial \Sigma_1}{\partial p^\mu} \\ &= -\frac{\mathrm{i}g^2}{32\pi^4 p^2}\int \mathrm{d}^4k\,\frac{p\cdot(p+k)}{(k^2-m_{\mathrm{ph}}^2+\mathrm{i}\epsilon)((p+k)^2-m_{\mathrm{ph}}^2+\mathrm{i}\epsilon)^2}.\end{aligned} \tag{26}$$

To evaluate it, use

$$\frac{1}{AB^n} = \int_0^1 \mathrm{d}x\,\frac{nx^{n-1}}{[xB+(1-x)A]^{n+1}}, \tag{27}$$

with

$$A = k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon, \qquad B = (p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon, \qquad n = 2. \tag{28}$$

Then

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= -\frac{\mathrm{i}g^2}{32\pi^4 p^2}\int_0^1 \mathrm{d}x\int \mathrm{d}^4k\,\frac{2x\,p\cdot(p+k)}{\left[k^2+2x\,p\cdot k+xp^2-m_{\mathrm{ph}}^2+\mathrm{i}\epsilon\right]^3} \\ &= \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x\int \mathrm{d}^4k\,\frac{x\,p\cdot(p+k)}{\left[m_{\mathrm{ph}}^2-k^2-2x\,p\cdot k-xp^2-\mathrm{i}\epsilon\right]^3}.\end{aligned} \tag{29}$$

Complete the square by shifting the loop momentum:

$$k' = k + xp, \qquad \Delta(x) = m_{\mathrm{ph}}^2 - p^2 x(1-x). \tag{30}$$

The numerator becomes

$$x\,p\cdot(p+k) = x\left[p^2(1-x) + p\cdot k'\right]. \tag{31}$$

The term proportional to $p\cdot k'$ is odd in the shifted integration variable and vanishes. Therefore

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = \frac{\mathrm{i}g^2}{16\pi^4}\int_0^1 \mathrm{d}x\,x(1-x)\int \mathrm{d}^4k'\,\frac{1}{[\Delta(x)-k'^2-\mathrm{i}\epsilon]^3}. \tag{32}$$

Using

$$\int \frac{\mathrm{d}^d l}{(2\pi)^d}\frac{1}{(l^2-\Delta+\mathrm{i}\epsilon)^m} = \frac{\mathrm{i}(-1)^m}{(4\pi)^{d/2}}\frac{\Gamma(m-d/2)}{\Gamma(m)}\Delta^{d/2-m}, \tag{33}$$

with $d = 4$ and $m = 3$, gives

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= -\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\frac{x(1-x)}{m_{\mathrm{ph}}^2-p^2x(1-x)} \\ &= \frac{g^2}{32\pi^2}\frac{\partial}{\partial p^2}\int_0^1 \mathrm{d}x\,\ln\left[m_{\mathrm{ph}}^2-p^2x(1-x)\right].\end{aligned} \tag{34}$$

Integrating back gives

$$\Sigma_{1R}(p^2) = \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\ln\left[m_{\mathrm{ph}}^2-p^2x(1-x)\right] + C. \tag{35}$$

The on-shell condition $\Sigma_{1R}(m_{\text{ph}}^2) = 0$ fixes the constant:

$$C = -\frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\text{ph}}^2 - m_{\text{ph}}^2 x(1-x)\right]. \tag{36}$$

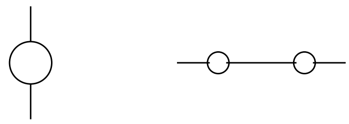
Thus

$$\Sigma_{1R}(p^2) = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[\frac{m_{\text{ph}}^2 - p^2 x(1-x)}{m_{\text{ph}}^2(1-x+x^2)}\right]. \tag{37}$$

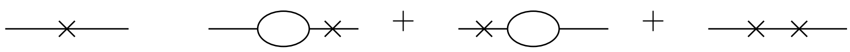
This expression is finite and vanishes at $p^2 = m_{\text{ph}}^2$ by construction.

At higher orders, the same logic must be applied to divergent subgraphs inside larger graphs. If a one-loop bubble appears as a subgraph, the renormalized graph is obtained by adding the corresponding local counterterm graph:

1. The original larger graph may contain a bubble subgraph:



2. Each divergent bubble must be accompanied by a counterterm insertion. The cross denotes the local counterterm:



3. The sum of the original graph and all required counterterm graphs is finite. In this sense the divergent bubble subgraph is replaced by its finite renormalized value $-\mathrm{i}\Sigma_{1R}$.

The important point is locality: whenever a divergent short-distance subgraph appears, its divergent part has the same form as a local counterterm with the same external legs.

1.3 Degree of divergence and required counterterms

The superficial degree of divergence is a quick estimate of how a Feynman integral behaves at large loop momentum. For the one-loop self-energy in d spacetime dimensions,

$$\Sigma_1(p^2, m^2; d) = \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k \, \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{38}$$

At large k, each propagator behaves as $1/k^2$, while the measure contributes $k^{d-1}\mathrm{d}k$. Therefore

$$\mathrm{d}^d k \, \frac{1}{k^2 k^2} \sim \mathrm{d}k \, k^{d-5}. \tag{39}$$

Equivalently, before the final radial integral, the graph has superficial degree

$$D = d - 4. \tag{40}$$

Thus

$$d < 4 \implies \text{the graph is superficially convergent,} \tag{41}$$
$$d = 4 \implies \text{the graph is logarithmically divergent,} \tag{42}$$
$$d > 4 \implies \text{the graph is power divergent by superficial counting.} \tag{43}$$

External-momentum derivatives reduce the degree of divergence. Differentiating once gives

$$\frac{\partial \Sigma_1}{\partial p^\mu} = -\frac{\mathrm{i}g^2}{(2\pi)^d} \int \mathrm{d}^d k \, \frac{(p+k)_\mu}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)^2}. \tag{44}$$

At large k,

$$\mathrm{d}^d k \, \frac{k}{k^2 (k^2)^2} \sim \mathrm{d}k \, k^{d-6}, \tag{45}$$

so the degree has been lowered from $d-4$ to $d-5$. If $d = 5$, naive power counting gives a logarithm. The leading logarithmic piece is odd under symmetric integration of the loop momentum, so it does not produce a Lorentz-invariant divergence in the self-energy.

Differentiating a second time gives

$$\frac{\partial^2 \Sigma_1}{\partial p^\mu \partial p^\nu} = \frac{\mathrm{i}g^2}{(2\pi)^d} \int \mathrm{d}^d k \, \frac{2(p+k)_\mu (p+k)_\nu - g_{\mu\nu}((p+k)^2 - m^2)}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)^3}. \tag{46}$$

Its superficial degree is

$$D_{\partial^2 \Sigma} = d - 6. \tag{47}$$

In four dimensions, this is finite. When we integrate twice to reconstruct Σ_1, the most general missing polynomial is

$$A + B_\mu p^\mu. \tag{48}$$

Since the self-energy is a Lorentz scalar, the linear term is forbidden, so $B_\mu = 0$. The only local ambiguity in $d = 4$ is the constant A, which is precisely a mass counterterm.

In six dimensions, the original graph has $D = 2$. After enough derivatives make the integral finite, integrating back allows a quadratic local polynomial:

$$A + Bp^2 + C_\mu p^\mu. \tag{49}$$

Lorentz invariance again sets $C_\mu = 0$. Therefore in $d = 6$ the two-point counterterm must contain both a mass counterterm and a wave-function counterterm. With the sign convention used below, their contribution to the self-energy is

$$\Sigma_{\mathrm{ct}}(p^2) = \delta m^2 - \delta Z \, p^2. \tag{50}$$

The corresponding local Lagrangian terms are

$$\mathcal{L}_{\mathrm{ct}} = -\frac{1}{2}\delta Z \, \partial_\mu \phi \, \partial^\mu \phi - \frac{1}{2}\delta m^2 \phi^2. \tag{51}$$

The δZ term is called wave-function or field-strength renormalization.

The field-strength factor relates the bare field to the renormalized field:

$$\phi_0 = Z^{1/2}\phi, \qquad \phi = Z^{-1/2}\phi_0. \tag{52}$$

Therefore

$$\begin{aligned} G_{2(0)}(p) &= \langle 0 | \, \mathrm{T}\{\phi_0(p)\phi_0(-p)\} \, | 0 \rangle \\ &= Z \, \langle 0 | \, \mathrm{T}\{\phi(p)\phi(-p)\} \, | 0 \rangle = Z G_2(p). \end{aligned} \tag{53}$$

The full bare propagator is

$$G_{2(0)}(p^2) = \frac{\mathrm{i}}{p^2 - m_0^2 - \Sigma_{1(0)}(p^2) + \mathrm{i}\epsilon}. \tag{54}$$

The physical mass is the pole position:

$$m_{\mathrm{ph}}^2 - m_0^2 - \Sigma_{1(0)}(m_{\mathrm{ph}}^2) = 0, \qquad \Sigma_{1(0)}(m_{\mathrm{ph}}^2) = m_{\mathrm{ph}}^2 - m_0^2. \tag{55}$$

Hence

$$p^2 - m_0^2 - \Sigma_{1(0)}(p^2) = p^2 - m_{\mathrm{ph}}^2 - \big[\Sigma_{1(0)}(p^2) - \Sigma_{1(0)}(m_{\mathrm{ph}}^2)\big]. \tag{56}$$

When only mass renormalization is needed, this motivates

$$\Sigma_{1R}(p^2) = \Sigma_{1(0)}(p^2) - \Sigma_{1(0)}(m_{\mathrm{ph}}^2). \tag{57}$$

With a wave-function counterterm included, the general form is

$$\Sigma_{1R}(p^2) = \Sigma_{1(0)}(p^2) + \delta m^2 - \delta Z\, p^2. \tag{58}$$

Then the bare and renormalized propagators are related by

$$G_{2(0)}(p^2) = \frac{\mathrm{i}Z}{p^2 - m_{\mathrm{ph}}^2 - \Sigma_{1R}(p^2) + \mathrm{i}\epsilon}. \tag{59}$$

The on-shell scheme imposes two conditions. The first fixes the pole:

$$\Sigma_{1R}(m_{\mathrm{ph}}^2) = 0. \tag{60}$$

The second fixes the pole residue. Expanding near $p^2 = m_{\mathrm{ph}}^2$,

$$\begin{aligned}\Sigma_{1R}(p^2) &= \Sigma_{1R}(m_{\mathrm{ph}}^2) + (p^2 - m_{\mathrm{ph}}^2)\Sigma'_{1R}(m_{\mathrm{ph}}^2) + O\big((p^2 - m_{\mathrm{ph}}^2)^2\big)\\ &= (p^2 - m_{\mathrm{ph}}^2)\Sigma'_{1R}(m_{\mathrm{ph}}^2) + O\big((p^2 - m_{\mathrm{ph}}^2)^2\big).\end{aligned} \tag{61}$$

Thus

$$G_2(p^2) \simeq \frac{\mathrm{i}}{(p^2 - m_{\mathrm{ph}}^2)\big[1 - \Sigma'_{1R}(m_{\mathrm{ph}}^2)\big]}. \tag{62}$$

Requiring unit residue gives

$$\Sigma'_{1R}(m_{\mathrm{ph}}^2) = 0. \tag{63}$$

For the bare propagator,

$$\begin{aligned}G_{2(0)}(p^2) &\simeq \frac{\mathrm{i}}{(p^2 - m_{\mathrm{ph}}^2)\Big[1 - \Sigma'_{1(0)}(m_{\mathrm{ph}}^2)\Big]}\\ &\equiv \frac{\mathrm{i}R_{(0)}}{p^2 - m_{\mathrm{ph}}^2},\end{aligned} \tag{64}$$

so

$$R_{(0)} = \frac{1}{1 - \Sigma'_{1(0)}(m_{\mathrm{ph}}^2)}. \tag{65}$$

Since $G_{2(0)} = ZG_2$, choosing the renormalized propagator to have unit residue means

$$Z = R_{(0)} = \frac{1}{1 - \Sigma'_{1(0)}(m_{\text{ph}}^2)}. \tag{66}$$

Finally, rewrite the bare ϕ^3 Lagrangian:

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi_0\, \partial^\mu \phi_0 - \frac{1}{2}m_0^2\phi_0^2 - \frac{g_0}{3!}\phi_0^3. \tag{67}$$

Using $\phi_0 = Z^{1/2}\phi$,

$$\mathcal{L} = \frac{1}{2}Z\, \partial_\mu \phi\, \partial^\mu \phi - \frac{1}{2}Zm_0^2\phi^2 - \frac{g_0}{3!}Z^{3/2}\phi^3. \tag{68}$$

Define

$$Z = 1 - \delta Z, \qquad Zm_0^2 = m_{\text{ph}}^2 + \delta m^2, \qquad Z^{3/2}g_0 = g_{\text{ph}} + \delta g. \tag{69}$$

Then

$$\begin{aligned}\mathcal{L} = {} & \frac{1}{2}\partial_\mu \phi\, \partial^\mu \phi - \frac{1}{2}m_{\text{ph}}^2\phi^2 - \frac{g_{\text{ph}}}{3!}\phi^3 \\ & - \frac{1}{2}\delta Z\, \partial_\mu \phi\, \partial^\mu \phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta g}{3!}\phi^3.\end{aligned} \tag{70}$$

The first line supplies the renormalized propagator and vertex. The second line supplies the counterterm vertices needed to cancel divergences order by order.

1.4 Arbitrariness in a renormalization graph

Counterterms are required by finiteness, but finiteness does not uniquely determine their finite parts. This freedom is the origin of different renormalization schemes.

In four-dimensional ϕ^3 theory at one loop, the two-point graph is logarithmically divergent, so a mass counterterm is enough and we may set

$$\delta Z = 0 \tag{71}$$

at this order. The renormalized self-energy can then be written as

$$\Sigma_{1R}(p^2) = \Sigma_{1,\text{fin}}(p^2) + \big(\Sigma_{1,\text{div}} + \delta m^2\big). \tag{72}$$

The divergent part of δm^2 must cancel $\Sigma_{1,\text{div}}$, but the finite part is still a convention.

In the mass-shell scheme, the convention is fixed by requiring the pole of the full propagator to be at $p^2 = m_{\text{ph}}^2$:

$$\Sigma_{1R}(m_{\text{ph}}^2) = 0. \tag{73}$$

Thus

$$0 = \Sigma_{1,\text{fin}}(m_{\text{ph}}^2) + \Sigma_{1,\text{div}} + \delta m^2, \tag{74}$$

and hence

$$\delta m_{\text{OS}}^2 = -\Sigma_{1,\text{div}} - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2). \tag{75}$$

The on-shell renormalized self-energy is

$$\Sigma_{1R}^{\text{OS}}(p^2) = \Sigma_{1,\text{fin}}(p^2) - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2). \tag{76}$$

Therefore, through order g^2,

$$G_2(p^2) = \frac{\mathrm{i}}{p^2 - m_{\mathrm{ph}}^2 - \Sigma_{1,\mathrm{fin}}(p^2) + \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4)}. \tag{77}$$

The bare mass is

$$m_0^2 = m_{\mathrm{ph}}^2 + \delta m_{\mathrm{OS}}^2 = m_{\mathrm{ph}}^2 - \Sigma_{1,\mathrm{div}} - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4). \tag{78}$$

Now choose a different finite part. The only requirement for finiteness is

$$\delta m^2 = -\Sigma_{1,\mathrm{div}} + \text{finite}. \tag{79}$$

For example, choose the finite part to be $-g^2 C$:

$$\delta m_C^2 = -\Sigma_{1,\mathrm{div}} - g^2 C, \tag{80}$$

where C is an arbitrary finite constant. The renormalized self-energy is then

$$\Sigma_{1R}^{(C)}(p^2, m^2) = \Sigma_{1,\mathrm{fin}}(p^2, m^2) - g^2 C. \tag{81}$$

Here m is the renormalized mass parameter in this scheme. It is not automatically the physical pole mass. The physical pole must be found from

$$p^2 - m^2 - \Sigma_{1R}^{(C)}(p^2, m^2) = 0 \tag{82}$$

order by order in perturbation theory. The propagator is

$$G_2^{(C)}(p^2) = \frac{\mathrm{i}}{p^2 - m^2 - \Sigma_{1,\mathrm{fin}}(p^2, m^2) + g^2 C + O(g^4)}. \tag{83}$$

The bare mass in this scheme is

$$m_0^2 = m^2 + \delta m_C^2 = m^2 - \Sigma_{1,\mathrm{div}} - g^2 C + O(g^4). \tag{84}$$

The bare mass is the same object in every scheme. Equating the on-shell expression and the C-scheme expression gives

$$m^2 - \Sigma_{1,\mathrm{div}} - g^2 C = m_{\mathrm{ph}}^2 - \Sigma_{1,\mathrm{div}} - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4). \tag{85}$$

The divergent pieces cancel, leaving

$$m^2 = m_{\mathrm{ph}}^2 + g^2 C - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4). \tag{86}$$

This is a finite redefinition between two renormalization schemes. The parameter m is scheme dependent, while m_{ph} is defined by the pole of the propagator and is observable. Changing the finite part of the counterterm changes the numerical value assigned to the renormalized parameter, but it does not change physical predictions when all quantities are converted consistently.

1.5 Dimensional regularization

Dimensional regularization is another systematic way to handle ultraviolet divergences. Instead of putting a hard cutoff Λ on the loop momentum, we temporarily continue the number of spacetime dimensions from 4 to a complex variable d. The integral is first evaluated in a region of d where it converges, and the result is then analytically continued back to the physical value. In this language, UV divergences appear as poles of Gamma functions.

For the same one-loop two-point function in ϕ^3 theory, we write

$$\Sigma_1(p^2, m^2; d) = \frac{\mathrm{i}g^2\mu^{4-d}}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{87}$$

The factor μ^{4-d} is inserted so that the coupling g keeps its four-dimensional mass dimension while d is moved away from 4.

The basic Gaussian integral is

$$\int \mathrm{d}^n x\, \mathrm{e}^{-x^2} = (\sqrt{\pi})^n = \pi^{n/2}. \tag{88}$$

After Wick rotation, $k^0 = \mathrm{i}k_E^0$, the Minkowski square becomes $k^2 = -k_E^2$, and the measure contributes one factor of i:

$$\begin{aligned}\int \mathrm{d}^d k\, \mathrm{e}^{k^2} &= \mathrm{i} \int \mathrm{d}^d k_E\, \mathrm{e}^{-k_E^2} \\ &= \mathrm{i}\left(\int \mathrm{d}k_E^0\, \mathrm{e}^{-(k_E^0)^2}\right)\left(\int \mathrm{d}^{d-1}\vec{k}_E\, \mathrm{e}^{-\vec{k}_E^2}\right) \\ &= \mathrm{i}\sqrt{\pi}\, \pi^{(d-1)/2} = \mathrm{i}\pi^{d/2}.\end{aligned} \tag{89}$$

One way to derive the dimensionally regularized answer is to use Schwinger parameters. The propagator can be represented as

$$\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} = \int_0^\infty \mathrm{d}a\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)}. \tag{90}$$

Using one parameter for each propagator gives

$$\begin{aligned}\Sigma_1 &= \frac{\mathrm{i}g^2\mu^{4-d}}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)} \mathrm{e}^{-\mathrm{i}b(m^2 - (p+k)^2 - \mathrm{i}\epsilon)} \\ &= \frac{\mathrm{i}g^2\mu^{4-d}}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}[(a+b)m^2 - (a+b)k^2 - bp^2 - 2b\,p\cdot k - \mathrm{i}\epsilon(a+b)]}.\end{aligned} \tag{91}$$

Complete the square by defining

$$k' = k + \frac{b}{a+b}p. \tag{92}$$

Then

$$(a+b)k^2 + 2b\,p \cdot k + bp^2 = (a+b)k'^2 + \frac{ab}{a+b}p^2. \tag{93}$$

It is convenient to change variables from (a, b) to

$$z = a + b, \qquad x = \frac{a}{a+b}, \qquad a = xz, \qquad b = (1-x)z. \tag{94}$$

The Jacobian is

$$\mathrm{d}a\,\mathrm{d}b = \begin{vmatrix} \frac{\partial a}{\partial x} & \frac{\partial a}{\partial z} \\ \frac{\partial b}{\partial x} & \frac{\partial b}{\partial z} \end{vmatrix} \mathrm{d}x\,\mathrm{d}z = \begin{vmatrix} z & x \\ -z & 1-x \end{vmatrix} \mathrm{d}x\,\mathrm{d}z = z\,\mathrm{d}x\,\mathrm{d}z. \tag{95}$$

Thus $z \in (0,\infty)$ and $x \in (0,1)$, and the exponent contains the standard combination

$$\Delta(x) = m^2 - p^2 x(1-x). \tag{96}$$

After the Gaussian loop integral and the z integral, one obtains

$$\Sigma_1(p^2, m^2; d) = -\frac{g^2\mu^{4-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\left[m^2 - p^2x(1-x)\right]^{d/2-2}. \tag{97}$$

The same result follows more directly from the Feynman-parameter identity

$$\frac{1}{AB} = \int_0^1 \mathrm{d}x\,\frac{1}{[xB+(1-x)A]^2}. \tag{98}$$

With

$$A = k^2 - m^2 + \mathrm{i}\epsilon, \qquad B = (p+k)^2 - m^2 + \mathrm{i}\epsilon, \tag{99}$$

we get

$$\begin{aligned} \Sigma_1 &= \frac{\mathrm{i}g^2\mu^{4-d}}{2}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{1}{[(k+xp)^2 - \Delta(x) + \mathrm{i}\epsilon]^2} \\ &= -\frac{g^2\mu^{4-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\Delta(x)^{d/2-2}. \end{aligned} \tag{100}$$

The divergence is now entirely in $\Gamma(2-d/2)$. Near $d=4$, set

$$d = 4 - \epsilon. \tag{101}$$

Then

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \tag{102}$$

and

$$\Delta(x)^{d/2-2} = \Delta(x)^{-\epsilon/2} = 1 - \frac{\epsilon}{2}\ln\Delta(x) + O(\epsilon^2). \tag{103}$$

Using also

$$\mu^\epsilon(4\pi)^{\epsilon/2} = 1 + \epsilon\ln\mu + \frac{\epsilon}{2}\ln 4\pi + O(\epsilon^2), \tag{104}$$

the unrenormalized self-energy becomes

$$\Sigma_1 = -\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + \ln\mu^2 - \ln\Delta(x)\right] + O(\epsilon). \tag{105}$$

1.6 Minimal subtraction scheme

As we introduced above, we may have different schemes to choose for renormalization. The on-shell subtraction is especially transparent in dimensional regularization. Define

$$\Sigma_{1R}^{\mathrm{OS}}(p^2) = \Sigma_1(p^2, m_{\mathrm{ph}}^2; d) - \Sigma_1(m_{\mathrm{ph}}^2, m_{\mathrm{ph}}^2; d). \tag{106}$$

The two terms have the same pole, so the pole cancels before taking $\epsilon \to 0$. Since

$$\Delta_p(x) = m_{\text{ph}}^2 - p^2 x(1-x), \qquad \Delta_{\text{OS}}(x) = m_{\text{ph}}^2 - m_{\text{ph}}^2 x(1-x) = m_{\text{ph}}^2(1 - x + x^2), \tag{107}$$

we obtain

$$\Sigma_{1R}^{\text{OS}}(p^2) = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln \frac{m_{\text{ph}}^2 - p^2 x(1-x)}{m_{\text{ph}}^2(1 - x + x^2)}. \tag{108}$$

This is exactly the finite on-shell result obtained earlier by differentiating with respect to p^2.

But on-shell scheme would have problems when we are dealing with the massless case. So we introduce the **Minimal Subtraction (MS)**, which is a different prescription. It subtracts only the pole part at the physical dimension. In $d = 4 - \epsilon$, the mass counterterm in the MS scheme is chosen as

$$\delta m_{\text{MS}}^2 = \frac{g^2}{32\pi^2} \frac{2}{\epsilon}. \tag{109}$$

Therefore

$$\begin{aligned} \Sigma_{1R}^{\text{MS}}(p^2) &= \Sigma_1(p^2) + \delta m_{\text{MS}}^2 \\ &= \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \left[\ln\left(\frac{\Delta(x)}{4\pi\mu^2} \right) + \gamma_E \right] + O(\epsilon). \end{aligned} \tag{110}$$

This expression is finite, but it is not the on-shell scheme result. In particular, $\Sigma_{1R}^{\text{MS}}(m^2)$ is not required to vanish. The MS scheme removes only the pole; the finite part is chosen by convention.

The scale μ appears because the mass dimension of the coupling changes when the dimension is continued. In d dimensions,

$$[\mathcal{L}] = [m]^d, \qquad [\phi] = [m]^{(d-2)/2}, \tag{111}$$

and the interaction

$$\frac{g}{3!}\phi^3 \tag{112}$$

implies

$$[g] = [m]^{(6-d)/2}. \tag{113}$$

Around four dimensions, g has dimension one, and writing $g_0 = \mu^{(4-d)/2} g$ keeps the renormalized coupling normalized as a four-dimensional parameter. Around six dimensions, where ϕ^3 is classically marginal, the corresponding continuation is

$$g_0 = \mu^{3-d/2}(g + \delta g). \tag{114}$$

Now consider the same two-point graph near six dimensions. Set

$$d = 6 - \epsilon, \qquad \Delta(x) = m^2 - p^2 x(1-x). \tag{115}$$

The dimensionally regularized result is

$$\Sigma_1 = -\frac{g^2 \mu^{6-d}}{2(4\pi)^{d/2}} \Gamma\left(2 - \frac{d}{2}\right) \int_0^1 \mathrm{d}x \, \Delta(x)^{d/2-2}. \tag{116}$$

Here

$$\Gamma\left(2 - \frac{d}{2}\right) = \Gamma\left(-1 + \frac{\epsilon}{2}\right) = -\frac{2}{\epsilon} + \gamma_E - 1 + O(\epsilon). \tag{117}$$

Expanding the remaining factors gives

$$\Sigma_1 = -\frac{g^2}{128\pi^3}\int_0^1 \mathrm{d}x\,\Delta(x)\left[-\frac{2}{\epsilon} + \gamma_E - 1 + \ln\left(\frac{\Delta(x)}{4\pi\mu^2}\right)\right] + O(\epsilon). \tag{118}$$

Since

$$\int_0^1 \mathrm{d}x\,\Delta(x) = m^2 - \frac{p^2}{6}, \tag{119}$$

the pole part is

$$\Sigma_{1,\mathrm{pole}} = \frac{g^2}{64\pi^3}\frac{1}{\epsilon}\left(m^2 - \frac{p^2}{6}\right). \tag{120}$$

Thus a mass counterterm alone is no longer sufficient: the divergence contains both a constant and a p^2 term. Writing

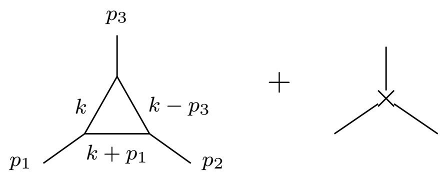
$$\Sigma_{1R} = \Sigma_1 + \delta m^2 - \delta Z\,p^2, \tag{121}$$

minimal subtraction at $d = 6 - \epsilon$ requires

$$\delta m^2 = -\frac{g^2 m^2}{64\pi^3}\frac{1}{\epsilon} = \frac{g^2 m^2}{64\pi^3}\frac{1}{d-6}, \tag{122}$$

$$\delta Z = -\frac{g^2}{6\cdot 64\pi^3}\frac{1}{\epsilon} = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6}. \tag{123}$$

In six dimensions the three-point vertex is also logarithmically divergent. The tree vertex is corrected by the one-loop triangle, and the divergence is canceled by a local three-leg counterterm:



To extract the UV pole, set the external momenta to zero after noting that the superficial divergence is logarithmic. Momentum-dependent terms are less divergent and affect only finite parts. The triangle contribution is then

$$\begin{aligned}\Gamma_1^{(3)} &= (-\mathrm{i}g\mu^{3-d/2})^3\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(k+p_1)^2 - m^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(k-p_2)^2 - m^2 + \mathrm{i}\epsilon}\\ &\xrightarrow{p_i\to 0} (-\mathrm{i}g\mu^{3-d/2})^3\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}^3}{(k^2 - m^2 + \mathrm{i}\epsilon)^3}\\ &= -\frac{\mathrm{i}g^3}{2(4\pi)^3}\left(\frac{2}{\epsilon} + \text{finite}\right).\end{aligned} \tag{124}$$

The counterterm vertex is $-\mathrm{i}\delta g$. Requiring the pole to cancel gives

$$-\mathrm{i}\delta g + \left[-\frac{\mathrm{i}g^3}{2(4\pi)^3}\frac{2}{\epsilon}\right] = 0, \tag{125}$$

so

$$\delta g = -\frac{g^3}{64\pi^3}\frac{1}{\epsilon}\,\mu^{3-d/2} = \frac{g^3}{64\pi^3}\frac{1}{d-6}\,\mu^{3-d/2}. \tag{126}$$

This is the vertex counterterm in minimal subtraction near six dimensions.

1.7 Two loop graph in massless ϕ^3 theory

The on-shell scheme is natural for a massive particle, because the physical mass is defined by an isolated pole at $p^2 = m_{\rm ph}^2$. For a strictly massless interacting theory this prescription becomes delicate. The reason is not a new ultraviolet problem; it is an infrared problem caused by taking the subtraction point to the massless pole.

Recall the one-loop on-shell mass counterterm in dimensional regularization $d = 4 - \epsilon$:

$$\delta m_{\rm OS}^2 = -\Sigma_1(m_{\rm ph}^2, m_{\rm ph}^2) = \frac{g^2\mu^\epsilon}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\left[m_{\rm ph}^2(1-x+x^2)\right]^{d/2-2}, \tag{127}$$

with the approximation formulas:

$$\Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \qquad A^{-\epsilon/2} = 1 - \frac{\epsilon}{2}\ln A + O(\epsilon^2). \tag{128}$$

These gives

$$\delta m_{\rm OS}^2 = \frac{g^2}{32\pi^2}\left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + \ln\mu^2 - \ln m_{\rm ph}^2 - \int_0^1 \mathrm{d}x\,\ln\bigl(1-x+x^2\bigr)\right] + O(\epsilon). \tag{129}$$

The pole $2/\epsilon$ is the ultraviolet divergence. The separate term

$$-\ln m_{\rm ph}^2 \tag{130}$$

blows up when $m_{\rm ph} \to 0$. This is an infrared divergence. It appears because the on-shell subtraction point has been moved to $p^2 = 0$, where a massless field has long-range propagation.

Minimal subtraction does not require an on-shell subtraction point. Setting $m = 0$ after MS subtraction gives

$$\begin{aligned}\Sigma_{1R}^{\rm MS}(p^2)\big|_{m=0} &= \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\left[\ln\left(\frac{-p^2x(1-x)}{4\pi\mu^2}\right) + \gamma_E\right]\\ &= \frac{g^2}{32\pi^2}\left[\ln\left(\frac{-p^2}{4\pi\mu^2}\right) + \gamma_E - 2\right].\end{aligned} \tag{131}$$

This expression is meaningful for $p^2 \neq 0$, but it is still singular as $p^2 \to 0$. Therefore the failure of the mass-shell scheme in the strictly massless case is an infrared statement: near the massless point, the full propagator is not described by the same simple isolated massive-particle pole.

For perturbation theory it is useful to separate the Lagrangian into the free part, the basic interaction, and the counterterms:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_b + \mathcal{L}_{ct} \tag{132}$$

with

$$\mathcal{L}_0 = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2, \tag{133}$$

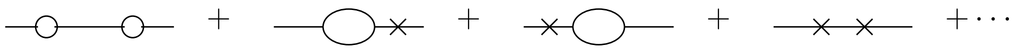
$$\mathcal{L}_b = -\frac{g}{3!}\phi^3, \tag{134}$$

and

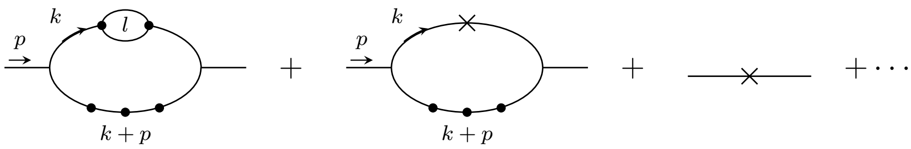
$$\mathcal{L}_{ct} = -\frac{1}{2}\delta Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta g}{3!}\phi^3 - \delta h\,\phi. \tag{135}$$

The term δh cancels tadpoles, namely one-point divergences. At order g^4 we must draw not only the genuine two-loop graphs. We must also draw graphs in which a lower-order divergent subgraph has already been replaced by its local counterterm. The relevant classes are:

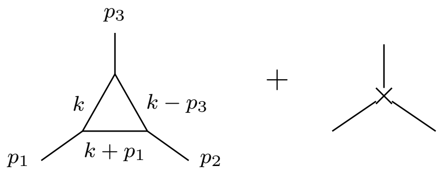
1. Iterated one-loop self-energy insertions on the same propagator line:



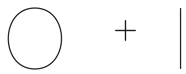
2. A two-loop self-energy graph with a one-loop two-point subdivergence:



3. A one-loop three-point vertex correction and its vertex counterterm:

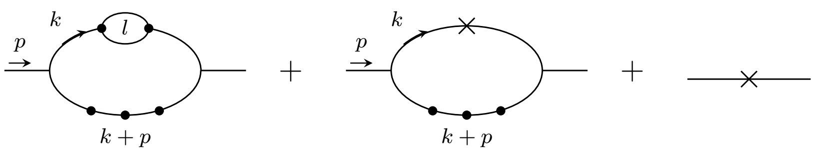


Each cross stands for a local counterterm. The rule is simple but important: every divergent subgraph needs its own subtraction before the remaining overall divergence can be removed. Tadpoles are not shown in the main list, because they are local one-point divergences and are canceled by the $\delta h\,\phi$ term:



A pole multiplying a non-polynomial function such as $\ln(-p^2)$ is called a non-local subdivergence. It cannot be canceled by a local term in the Lagrangian. After all subdivergences have been subtracted, the remaining overall divergence is local: in momentum space it is a polynomial in the external momenta.

Now we consider the two-loop self-energy in $d = 6$. The diagram that matters for subdivergences is the graph in which a one-loop two-point subgraph is inserted into an internal line. We must include three contributions:



(a) (b) (c)

Graph (a) is the genuine two-loop graph. Graph (b) replaces the divergent one-loop subgraph inside (a) by the lower-order counterterm. Graph (c) is the remaining overall local two-point counterterm. For a two-point function the cross abbreviates the local insertion $\delta m^2 - \delta Z\,p^2$; in the massless calculation below the relevant overall counterterm is the wave-function part. The sum of all three graphs is finite.

$$\begin{aligned}\Sigma_{2a} = (\mathrm{i}g\mu^{3-d/2})^4 \frac{1}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{d}^d l}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{l^2-m^2+\mathrm{i}\epsilon} \\ \times \frac{\mathrm{i}}{(k-l)^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(k+p)^2-m^2+\mathrm{i}\epsilon}. \qquad (136)\end{aligned}$$

The two factors of $1/(k^2-m^2)$ are the two propagator segments on the line that contains the one-loop subgraph. For convenience, we now take the massless case:

$$\Sigma_{2a} = \frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{d}^dl}{(2\pi)^d}\frac{1}{(k^2+\mathrm{i}\epsilon)^2(l^2+\mathrm{i}\epsilon)((k-l)^2+\mathrm{i}\epsilon)((k+p)^2+\mathrm{i}\epsilon)}. \tag{137}$$

After Wick rotation, we have

$$\begin{aligned}
\Sigma_{2a} &= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{\mathrm{d}^dl_E}{(2\pi)^d}\frac{1}{[l_E^2(k_E-l_E)^2][(k_E^2)^2(k_E+p_E)^2]}\\
&= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{(k_E^2)^2(k_E+p_E)^2}\left(\int \frac{\mathrm{d}^dl_E}{(2\pi)^d}\frac{1}{l_E^2(k_E-l_E)^2}\right)\\
&= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{(k_E^2)^2(k_E+p_E)^2}\mathrm{i}\pi^{d/2}\frac{\Gamma(2-d/2)}{(2\pi)^d}\int_0^1 \mathrm{d}x\,[-k_E^2x(1-x)]^{d/2-2}\\
&= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{(k_E^2)^2(k_E+p_E)^2}\mathrm{i}\pi^{d/2}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(-k_E^2)^{d/2-2}\\
&= \frac{\mathrm{i}g^4}{2^{13}\pi^9}(16\pi^3\mu^4)^{3-d/2}\frac{\Gamma(2-d/2)\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}\int \mathrm{d}^dk_E\,\frac{1}{(-k_E^2)^{4-d/2}(k_E+p_E)^2}.
\end{aligned} \tag{138}$$

For the integral, one can introduce a Feynman parameter

$$\frac{1}{A^{4-d/2}B} = \frac{\Gamma(5-d/2)}{\Gamma(4-d/2)}\int_0^1 \mathrm{d}x\,\frac{x^{3-d/2}}{[Ax+B(1-x)]^{5-d/2}}. \tag{139}$$

Thus, the result is

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\frac{\Gamma(2-d/2)\Gamma(5-d)\Gamma(\frac{d}{2}-1)^3\Gamma(d-4)}{\Gamma(d-2)\Gamma(4-d/2)\Gamma(3d/2-5)} \tag{140}$$

$$\equiv \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\Gamma(2-d/2)\Gamma(5-d)K(d) \tag{141}$$

Here $\Gamma(2-d/2)$ is the pole of the one-loop subgraph, while $\Gamma(5-d)$ is the remaining overall pole. The massless formula also has an infrared singularity as $p^2\to 0$; this is separate from the ultraviolet subdivergence.

Now expand near six dimensions. To keep track of the non-local momentum dependence, define

$$d = 6-\epsilon, \qquad L_p = \ln\left(\frac{-p^2}{4\pi\mu^2}\right). \tag{142}$$

The singular factors are

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(-1+\frac{\epsilon}{2}\right) = -\frac{2}{\epsilon}+O(1), \tag{143}$$

$$\Gamma(5-d) = \Gamma(-1+\epsilon) = -\frac{1}{\epsilon}+O(1), \tag{144}$$

$$\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6} = 1-\epsilon L_p+\frac{\epsilon^2}{2}L_p^2+O(\epsilon^3). \tag{145}$$

Therefore graph (a) has the pole structure

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{1}{(d-6)^2}+\frac{L_p+c_a}{d-6}+\frac{1}{2}L_p^2+c_a'L_p+c_a''+O(d-6)\right]. \tag{146}$$

The constants c_a, c'_a, c''_a are not important for the present point. The important term is $L_p/(d-6)$: it is a pole multiplying a non-polynomial function of the external momentum. A local counterterm cannot cancel such a term directly. It must be canceled by graph (b), which subtracts the divergent one-loop subgraph before the remaining overall divergence is removed.

For $m = 0$, the one-loop two-point counterterm in the subgraph is the wave-function counterterm

$$\delta_1 Z = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6} + O(g^4). \tag{147}$$

Because the counterterm in graph (b) sits on the line carrying momentum k, its Feynman rule contributes a factor proportional to $\delta_1 Z\, k^2$:

$$\begin{aligned}
\Sigma_{2b} &= (-\mathrm{i}g\mu^{3-d/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+p)^2-m^2}\delta_1 Z\cdot k^2 && (148)\\
&= \mathrm{i}g^2\mu^{6-d}\delta_1 Z\int\frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{k_E^2}{(k_E^2)^2(k_E+p_E)^2} && (149)\\
&= \mathrm{i}g^2\mu^{6-d}\delta_1 Z\frac{\mathrm{i}\pi^{d/2}}{(2\pi)^d}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(p^2)^{d/2-2} && (150)\\
&= \left(\frac{g^2}{64\pi^3}\right)\left(\frac{p^2}{6}\right)\frac{\Gamma(2-d/2)\Gamma^2(\frac{d}{2}-1)}{(d-6)\Gamma(d-2)}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d/2-3} && (151)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left(-\frac{2}{\epsilon}+\gamma_E-1\right)\frac{1}{\epsilon}\left(1-\frac{\epsilon}{2}L_p+\frac{\epsilon^2}{8}L_p^2\right) && (152)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left[-\frac{2}{\epsilon^2}+\frac{\gamma_E-1+L_p}{\epsilon}-\frac{1}{2}(\gamma_E-1)L_p-\frac{1}{4}L_p^2\right] && (153)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[-\frac{2}{(d-6)^2}+\frac{-L_p+\text{const.}}{d-6}-\frac{1}{4}L_p^2+\text{const.}\,L_p+O(d-6)\right]. && (154)
\end{aligned}$$

Adding graphs (a) and (b), the pole multiplying L_p cancels:

$$\begin{aligned}
&\Sigma_{2a}+\Sigma_{2b}\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{-1}{(d-6)^2}+\frac{1}{d-6}\text{Const.}+\frac{1}{4}\ln^2\frac{-p^2}{4\pi\mu^2}+\text{Const.}\ln\frac{-p^2}{4\pi\mu^2}+\text{const.}+O(d-6)\right]
\end{aligned} \tag{155}$$

The non-local pole has disappeared:

$$\frac{1}{d-6}\left(\ln\frac{-p^2}{4\pi\mu^2}+\text{Const.}\right)\to\frac{1}{d-6}\text{const.} \tag{156}$$

If this term survived, canceling it would require a non-local counterterm schematically of the form

$$\phi(x)\ln(\Box)\phi(x) \tag{157}$$

for the subdivergence $\ln(p^2)/(d-6)$, which is unacceptable in a local renormalizable Lagrangian. Since the non-local pole has been canceled by graph (b), the remaining divergent part is local and can be canceled by graph (c). The two-loop wave-function counterterm is fixed by

$$-p^2\delta_2 Z = -(\Sigma_{2a}+\Sigma_{2b})_{\text{local pole}}. \tag{158}$$

Equivalently, in MS notation,

$$\delta_2 Z = \left(\frac{g^2}{64\pi^3}\right)^2 \frac{1}{36}\left[-\frac{1}{(d-6)^2} + \frac{c_Z}{d-6}\right], \tag{159}$$

where c_Z depends on the precise subtraction convention.

Thus, the renormalized result is

$$\begin{aligned}\Sigma_{2R}^{\mathrm{MS}} &= \Sigma_{2a} + \Sigma_{2b} - p^2\delta_2 Z \\ &= \left(\frac{g^2}{64\pi^3}\right)^2 \frac{p^2}{36}\left[\frac{1}{4}L_p^2 + c_1 L_p + c_0\right].\end{aligned} \tag{160}$$

1.8 External-momentum derivatives and locality of counterterms

We now explain why the remaining overall counterterm of a 1PI graph must be a polynomial in the external momenta. This is the basic locality statement behind renormalization: after all subdivergences have been subtracted, the only ultraviolet divergence left in a graph is local.

For the two-loop graph above in $d = 6$, the superficial degree of the overall divergence is $D = 2$. Each derivative with respect to the external momentum lowers the degree by one. Therefore three external-momentum derivatives should remove the overall divergence. To make the bookkeeping clean, use the shorthand

$$\partial^p_\mu \equiv \frac{\partial}{\partial p^\mu}. \tag{161}$$

We also isolate the one-loop subgraph as

$$\Pi(k) = \int \mathrm{d}^d l\, \frac{1}{l^2(k-l)^2}. \tag{162}$$

Then the two-loop graph has the schematic form

$$\Sigma_{2a}(p) \sim \int \mathrm{d}^d k\, \Pi(k)\, \frac{1}{(k^2)^2\,(k+p)^2}, \tag{163}$$

$$\partial^p_\mu\partial^p_\nu\partial^p_\rho\Sigma_{2a}(p) \sim \int \mathrm{d}^d k\, \Pi(k)\, \partial^p_\mu\partial^p_\nu\partial^p_\rho\left[\frac{1}{(k^2)^2\,(k+p)^2}\right]. \tag{164}$$

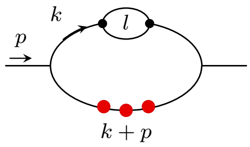
The derivative acts only on propagators through which the external momentum flows. In the graph above this is the line with momentum $k + p$:

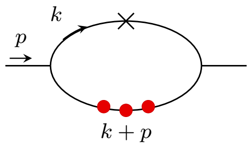
$$\partial^p_\mu \frac{1}{(k+p)^2 - m^2} = -\frac{2(k+p)_\mu}{[(k+p)^2 - m^2]^2}. \tag{165}$$

Thus differentiating a propagator produces one extra propagator power and a numerator proportional to the momentum through that line. In graph language we denote this bookkeeping operation by adding a marked insertion on the differentiated line:

$$\partial^p_\mu(\underline{\qquad}) \quad\longrightarrow\quad \underline{\qquad}. \tag{166}$$

The three red dots below mean that the external-momentum dependent propagator has been differentiated three times. They are not new interaction vertices; they only remind us where the powers of momentum in the numerator come from. The differentiated versions of graphs (a) and (b) are shown below.

(a''') 

(b''') 

The claim is slightly subtle. The graph (a''') by itself still contains the same one-loop subdivergence in the l loop. The finite object is the subtracted combination

$$(a''') + (b'''). \tag{167}$$

After this subtraction, the third derivative of the full graph is ultraviolet finite. Equivalently, the divergent part of the un-differentiated graph must obey

$$\partial^p_\mu \partial^p_\nu \partial^p_\rho \Sigma_{2,\mathrm{div}}(p) = 0. \tag{168}$$

Therefore $\Sigma_{2,\mathrm{div}}(p)$ can be at most a quadratic polynomial in the external momentum:

$$\Sigma_{2,\mathrm{div}}(p) = A(d)p^2 + B_\mu(d)p^\mu + C(d), \tag{169}$$

$$B_\mu(d) = 0 \qquad \text{by Lorentz invariance,} \tag{170}$$

$$\Sigma_{2,\mathrm{div}}(p) = A(d)p^2 + C(d). \tag{171}$$

The term $A(d)p^2$ is canceled by a wave-function counterterm. The constant $C(d)$ is a mass counterterm; in the massless MS example discussed above, the displayed overall divergence is proportional to p^2, so the relevant graph (c) is the δZ counterterm.

Let us check explicitly why the subtracted third derivative is finite. There are three ultraviolet regions to consider. If k and l become large together, the original overall degree is $D = 2$, and three derivatives lower it to $D = -1$, so this region is convergent. If $l \to \infty$ while k is fixed, the divergence is precisely the one-loop subgraph and is canceled by (b'''). The only region that is easy to miss is the nested region

$$l \gg k \gg |p|. \tag{172}$$

In this region the loop momentum l is much larger than the momentum k entering the subgraph, so the inner loop can be expanded in powers of k/l. Define

$$I(k) = \int \mathrm{d}^6 l \, \frac{1}{l^2 (l+k)^2}. \tag{173}$$

Here we keep $d = 6$, because this is the case where the two-loop graph has the subdivergence we want to subtract. The useful expansion is

$$(l+k)^2 = l^2 + 2l\cdot k + k^2 = l^2 \left(1 + \frac{2l\cdot k + k^2}{l^2}\right), \tag{174}$$

$$\frac{1}{(l+k)^2} = \frac{1}{l^2}\left[1 - \frac{2l\cdot k + k^2}{l^2} + \left(\frac{2l\cdot k + k^2}{l^2}\right)^2 + \cdots\right]. \tag{175}$$

This is the same as the Taylor expansion of the subgraph in its external momentum k:

$$I(k) = I(0) + k^\mu \left.\frac{\partial I}{\partial k^\mu}\right|_{k=0} + \frac{1}{2} k^\mu k^\nu \left.\frac{\partial^2 I}{\partial k^\mu \partial k^\nu}\right|_{k=0} + \cdots. \tag{176}$$

Now look term by term. In six dimensions $\mathrm{d}^6 l = \Omega_5 l^5 \mathrm{d}l$, and angular symmetry gives $\langle (l\cdot k)^2\rangle = l^2k^2/6$. The first three terms behave as

$$I_{(0)}(k) = \int \mathrm{d}^6 l\, \frac{1}{l^4} \sim \Omega_5 \int^\Lambda \mathrm{d}l\, l \sim \Lambda^2, \tag{177}$$

$$I_{(1)}(k) = -\int \mathrm{d}^6 l\, \frac{2l\cdot k}{l^6} = -2k_\mu \int \mathrm{d}^6 l\, \frac{l^\mu}{l^6} = 0, \tag{178}$$

$$\begin{aligned} I_{(2)}(k) &= \int \mathrm{d}^6 l \left(-\frac{k^2}{l^6} + \frac{4(l\cdot k)^2}{l^8} \right) \\ &\sim \Omega_5 \int^\Lambda \mathrm{d}l\, l^5 \left(-\frac{k^2}{l^6} + \frac{4}{l^8}\frac{l^2k^2}{6} \right) \\ &= -\frac{\Omega_5 k^2}{3} \int^\Lambda \frac{\mathrm{d}l}{l} \sim -\frac{\Omega_5 k^2}{3} \ln \Lambda. \end{aligned} \tag{179}$$

Thus $I_{(0)}$ is quadratically divergent, $I_{(1)}$ vanishes by oddness, and $I_{(2)}$ is logarithmically divergent.

Thus the divergent part of the subgraph is local in k. More explicitly, the divergent part can be written as a polynomial in k; the appearance of that polynomial depends on the regulator:

$$\begin{aligned} I^{(\Lambda)}_{\text{div}}(k) &= a_0\Lambda^2 + a_2 k^2 \ln \frac{\Lambda^2}{\mu^2}, && \text{hard cutoff,} \\ I^{(\text{DR})}_{\text{div}}(k) &= \frac{b_2 k^2}{d-6}, && \text{dimensional regularization near } d=6. \end{aligned} \tag{180}$$

The constants a_0, a_2, b_2 are regulator dependent. What matters is not their numerical value here, but the fact that the divergent part is a local polynomial: a constant plus a term proportional to k^2. The counterterm in graph (b) removes exactly these local divergent pieces. After subtraction, the large-k behavior of the inner subgraph is at worst

$$I_R(k) = I(k) - I_{\text{ct}}(k) = O\big(k^2 \ln k^2\big). \tag{181}$$

Terms of order k^3 and higher in the expansion of the subgraph are already finite in the l loop, so they do not introduce new subdivergences.

Finally combine this with the differentiated outer line. For $k \gg |p|$,

$$\partial^p_\mu \partial^p_\nu \partial^p_\rho \frac{1}{(k+p)^2} = O\left(\frac{1}{k^5}\right). \tag{182}$$

The worst large-k behavior of the subtracted differentiated two-loop graph is therefore obtained by multiplying the six-dimensional measure, the renormalized subgraph, the two k^2 propagators on the upper line, and the differentiated $k+p$ propagator:

$$\int^\infty \mathrm{d}k\, k^5 \left(k^2 \ln k^2\right) \frac{1}{(k^2)^2}\frac{1}{k^5} \sim \int^\infty \mathrm{d}k\, \frac{\ln k}{k^2} < \infty. \tag{183}$$

which is finite. This is the concrete reason why the non-local subdivergence must cancel between graphs (a) and (b), leaving only a local polynomial counterterm.

1.9 Composite operators

The new object in this subsection is a product of fields at nearby points, for example $\phi(x)\phi(y)$ when $x-y$ is small. Such a product is more singular than an ordinary field, because two quantum fields are being multiplied at almost the same spacetime point. The useful statement is that, at short distances, this product can be replaced by a sum of local operators at one point:

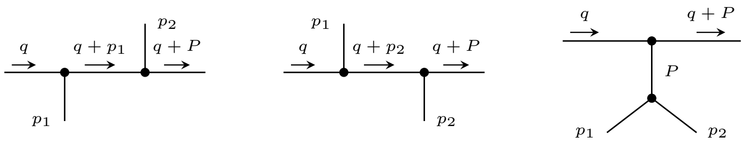
$$\phi(x)\phi(y) \sim C_1(x-y)\mathbb{1}+C_{\phi^2}(x-y)\left[\frac{1}{2}\phi^2(y)\right]+C_\phi(x-y)[\phi(y)]+C^\mu_{\partial\phi}(x-y)[\partial_\mu\phi(y)]+\dots \quad (184)$$

This is the operator product expansion, or OPE. The bracket $[A]$ means that the local operator has been renormalized. The functions C_i are ordinary functions, not operators; they are called Wilson coefficients. A Wilson coefficient contains the short-distance, high-momentum part of a graph, while the matrix element of the local operator contains the soft long-distance part.

We now see this mechanism in ϕ^3 theory. The large momentum q flows through the pair of fields that are being brought close together, whereas p_1 and p_2 are held fixed. In the labels of the figures,

$$P = p_1 + p_2, \qquad q^2 \gg p_1^2,\ p_2^2,\ P^2,\ m^2. \quad (185)$$

The relevant contributions to the four-point function are



(a) hard (b) hard (c) soft

The words "hard" and "soft" only describe which subgraph carries the large momentum. In graphs (a) and (b), the three propagators along the upper line are all hard. In graph (c), only the two upper propagators are hard; the lower part, which carries momentum P, remains soft and will be identified with a matrix element of a local operator.

The expansion used repeatedly below is the large-q expansion of a propagator. If p is fixed while q is taken large, then

$$\frac{1}{(q+p)^2-m^2} = \frac{1}{q^2}\left[1-\frac{2q\cdot p+p^2-m^2}{q^2}+O\left(\frac{1}{q^2}\right)\right]. \quad (186)$$

Equivalently, the small parameters are p/q and m/q. Only propagators carrying the hard momentum are expanded. The soft denominators are kept exact:

$$D_1 = p_1^2-m^2, \qquad D_2 = p_2^2-m^2, \qquad D_P = P^2-m^2. \quad (187)$$

$$(a) = (-\mathrm{i}g)^2\frac{\mathrm{i}}{q^2-m^2}\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{(q+p_1)^2-m^2}\frac{\mathrm{i}}{(q+P)^2-m^2}\frac{\mathrm{i}}{D_2} \quad (188)$$

$$\xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}+O\left(\frac{1}{q^7}\right), \quad (189)$$

$$(b) \xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}+O\left(\frac{1}{q^7}\right), \quad (190)$$

$$(a) + (b) \simeq 2\mathrm{i}g^2 \left(\frac{\mathrm{i}}{D_1} \frac{\mathrm{i}}{D_2} \right) \frac{1}{(q^2)^3}, \tag{191}$$

$$(c) = (-\mathrm{i}g)^2 \frac{\mathrm{i}}{q^2 - m^2} \frac{\mathrm{i}}{(q+P)^2 - m^2} \frac{\mathrm{i}}{D_P} \frac{\mathrm{i}}{D_1} \frac{\mathrm{i}}{D_2}. \tag{192}$$

For graph (c), the hard expansion contains two hard propagators instead of three:

$$\begin{aligned} (c) = &-\frac{\mathrm{i}g^2}{(q^2)^2} \left[1 - \frac{2q \cdot P}{q^2} + \frac{2m^2 - P^2}{q^2} + \frac{4(q \cdot P)^2}{(q^2)^2} + \cdots \right] \\ &\times \frac{1}{(P^2 - m^2)(p_1^2 - m^2)(p_2^2 - m^2)}, \qquad P = p_1 + p_2. \end{aligned} \tag{193}$$

The first important point is that the soft factor in (a)+(b) is exactly the lowest-order matrix element of the local operator $\frac{1}{2}\phi^2(0)$:

$$\begin{aligned} M_{\phi^2}(p_1, p_2) &\equiv \langle 0 | \, \mathrm{T}() \{ \tilde{\phi}(p_1) \tilde{\phi}(p_2) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle_{0,\mathrm{conn}} \\ &= \frac{\mathrm{i}}{p_1^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{p_2^2 - m^2 + \mathrm{i}\epsilon}. \end{aligned} \tag{194}$$

The factor $\frac{1}{2}$ removes the double counting of the two identical contractions. In the connected part,

$$\begin{aligned} &\langle 0 | \, \mathrm{T}() \{ \phi(x) \phi(y) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle_{0,\mathrm{conn}} \\ &\quad = \frac{1}{2} D(x) D(y) + \frac{1}{2} D(y) D(x) = D(x) D(y). \end{aligned} \tag{195}$$

The contraction in which the two fields inside $\phi^2(0)$ contract with each other gives a tadpole proportional to $D(0)$; it is not part of the connected graph displayed here and is removed by operator renormalization. In momentum space:

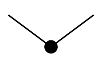
$$\begin{aligned} &\int \mathrm{d}^4 x \, \mathrm{d}^4 y \, \mathrm{e}^{-\mathrm{i}p_1 \cdot x - \mathrm{i}p_2 \cdot y} D(x) D(y) \\ &\quad = \frac{\mathrm{i}}{p_1^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{p_2^2 - m^2 + \mathrm{i}\epsilon}, \end{aligned} \tag{196}$$

$$M_{\phi^2}(p_1, p_2) = \int \mathrm{d}^4 x \, \mathrm{d}^4 y \, \mathrm{e}^{-\mathrm{i}p_1 \cdot x - \mathrm{i}p_2 \cdot y} \langle 0 | \, \mathrm{T}() \{ \phi(x) \phi(y) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle_{0,\mathrm{conn}} . \tag{197}$$

Here $\phi(0)$ is not Fourier transformed, because it is the local operator inserted at the origin. The same rule follows directly from the path integral:

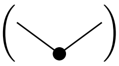
$$\langle 0 | \, \mathrm{T}() \{ \phi(x) \phi(y) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle = N \int [\mathrm{d}A] \, A(x) A(y) \frac{1}{2} A^2(0) \mathrm{e}^{\mathrm{i}S[A]}. \tag{198}$$

Here ϕ denotes the quantum operator, while A is the classical field inside the path integral. At lowest order, the insertion of $\frac{1}{2}\phi^2(0)$ therefore gives one local black-dot vertex with two scalar lines:



This insertion $\frac{1}{2}\phi^2(0)$ is our first example of a composite operator. Therefore

$$
\begin{aligned}
(a)+(b) &\simeq \underbrace{\frac{2\mathrm{i}g^2}{(q^2)^3}}_{C_{\phi^2}(q)} \underbrace{M_{\phi^2}(p_1,p_2)}_{\text{soft matrix element}} \\
&= \frac{2\mathrm{i}g^2}{(q^2)^3} \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\mathrm{conn}}
\end{aligned}
$$

$$= \frac{2\mathrm{i}g^2}{(q^2)^3}\Big($$  $$\Big). \tag{199}$$

Graph (c) is slightly different. The soft part is not the matrix element of $\frac{1}{2}\phi^2$; it is the matrix element of the elementary field ϕ between the vacuum and two external scalar legs. This matrix element first appears at order g:

$$
\begin{aligned}
&\langle 0|\,\mathrm{T}()\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle_{O(g)} \\
&\quad= \langle 0|\,\mathrm{T}()\{\phi(x)\phi(y)\phi(z)\int \mathrm{d}^4w\, \frac{-\mathrm{i}g}{3!}\phi^3(w)\}\,|0\rangle_0 \\
&\quad= -\mathrm{i}g\int \mathrm{d}^4w\, D(x-w)D(y-w)D(z-w).
\end{aligned}
\tag{200}
$$

We use

$$D(u-v) = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\mathrm{e}^{\mathrm{i}k\cdot(u-v)}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}. \tag{201}$$

In momentum space, with $P = p_1+p_2$, define

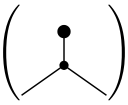
$$
\begin{aligned}
M_\phi(x;p_1,p_2) &\equiv \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\,|0\rangle_{O(g)} \\
&= \int \mathrm{d}^4y\mathrm{d}^4z\, \mathrm{e}^{-\mathrm{i}p_1\cdot y-\mathrm{i}p_2\cdot z}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}P\cdot w}D(x-w) \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}\,\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2-m^2+\mathrm{i}\epsilon}\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}(P+k)\cdot w} \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\frac{\mathrm{i}}{P^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{-\mathrm{i}P\cdot x}.
\end{aligned}
\tag{202}
$$

The factor $\mathrm{e}^{-\mathrm{i}P\cdot x}$ is the important part. Once the soft subgraph is represented by a local operator at $x=0$, powers of P in the hard coefficient become derivatives acting on that local operator:

$$\partial_\mu M_\phi(x;p_1,p_2)\Big|_{x=0} = -\mathrm{i}P_\mu M_\phi(0;p_1,p_2), \qquad P_\mu \to \mathrm{i}\partial_\mu, \qquad P^2 \to -\Box. \tag{203}$$

$$(c) \sim \frac{\mathrm{i}g}{(q^2)^2}\left[1-\frac{2q\cdot P}{q^2}+\frac{2m^2-P^2}{q^2}+\frac{4(q\cdot P)^2}{(q^2)^2}+\cdots\right]M_\phi(0;p_1,p_2) \tag{204}$$

$$= \frac{\mathrm{i}g}{(q^2)^2}\left[1-\frac{2\mathrm{i}q^\mu\partial_\mu}{q^2}+\frac{2m^2+\Box}{q^2}-\frac{4q^\mu q^\nu\partial_\mu\partial_\nu}{(q^2)^2}+\cdots\right]\langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\,|0\rangle\Bigg|_{x=0} \tag{205}$$

$$= \frac{\mathrm{i}g}{(q^2)^2}\sum_n c_n(q,\partial)\Bigg($$  $$\Bigg). \tag{206}$$

Putting the two types of terms together gives the OPE form of this four-point function:

$$\begin{aligned}&\langle 0|\,\mathrm{T}()\{\tilde{\phi}(q)\phi(0)\tilde{\phi}(p_1)\tilde{\phi}(p_2)\}\,|0\rangle\\ &\quad\sim \sum_i C_i(q)\,\langle 0|\,\mathrm{T}()\{O_i(0)\tilde{\phi}(p_1)\tilde{\phi}(p_2)\}\,|0\rangle\,.\end{aligned} \tag{207}$$

In this example the first few local operators are

$$O_1 = \frac{1}{2}\phi^2, \qquad \text{from graphs (a) and (b),} \tag{208}$$

$$O_2 = \phi, \qquad \text{from the leading term of graph (c),} \tag{209}$$

$$O_3 = \partial_\mu\phi,\ \Box\phi, \dots, \qquad \text{from higher terms in graph (c).} \tag{210}$$

The lesson is the separation of scales: the large momentum q is stored in the Wilson coefficients $C_i(q)$, while the dependence on the soft external momenta is stored in matrix elements of local operators.

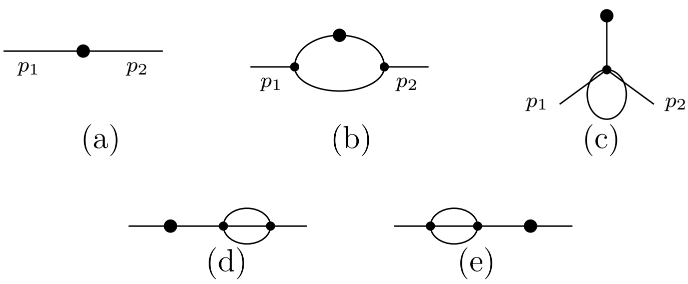
1.10 Renormalization of composite operators

The previous subsection used $\frac{1}{2}\phi^2(0)$ as a local insertion. This is not yet a complete definition of the operator. When the fields inside ϕ^2 approach interaction vertices, new ultraviolet divergences can occur at the insertion point itself. These divergences are not removed by renormalizing only the parameters in the Lagrangian; they require a renormalization of the composite operator.

Consider ϕ^3 theory in $d = 6 - \epsilon$ dimensions and the Green function with one insertion of $\frac{1}{2}\phi^2$:

$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle \tag{211}$$

The connected graphs up to order g^2 are shown below. The black dot is the insertion of $\frac{1}{2}\phi^2$. A crossed mark, when it appears later, denotes a local counterterm.



We now Fourier transform the two ordinary external fields, while the composite operator remains localized at the point z:

$$\begin{aligned}G &= \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(z)\}\,|0\rangle\\ &= \int \mathrm{d}^d x\,\mathrm{d}^d y\,\mathrm{e}^{\mathrm{i}(p_1\cdot x + p_2\cdot y)}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle\,.\end{aligned} \tag{212}$$

The lowest-order graph (a) is just the insertion vertex connected to the two external propagators. We use the shorthand

$$S_i = \frac{\mathrm{i}}{p_i^2 - m^2 + \mathrm{i}\epsilon}, \qquad S_{12} = S_1 S_2. \tag{213}$$

$$G_a = S_{12}. \tag{214}$$

For the order-g^2 graphs, the large-k behavior is

$$G_b \sim \int \mathrm{d}^6k \, \frac{1}{k^2}\frac{1}{k^2}\frac{1}{k^2} \sim \int^\Lambda \frac{\mathrm{d}k}{k}, \quad \text{logarithmically divergent,} \tag{215}$$

$$G_c,\ G_d,\ G_e \sim \int \mathrm{d}^6k \, \frac{1}{k^4} \sim \int^\Lambda \mathrm{d}k\, k, \qquad \text{quadratically divergent by power counting.} \tag{216}$$

Graphs (d) and (e) are ordinary self-energy corrections on the external lines. Their divergences are cancelled by the usual wave-function and mass counterterms. After ordinary field and parameter renormalization has been performed, they do not give a new counterterm for $[\phi^2]$.

Graphs (b) and (c) are different. Their short-distance divergence sits at the composite-operator insertion itself, so the bare operator ϕ^2 is not enough. We must add local counterterms to define the finite operator $[\phi^2]$.

$$G_b = S_{12}(-\mathrm{i}g)^2 \int \frac{\mathrm{d}^dk}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_1-k)^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_2+k)^2 - m^2 + \mathrm{i}\epsilon}. \tag{217}$$

After the continuation $g \to g\mu^{(6-d)/2}$, the loop part is

$$\begin{aligned} G_b &= S_{12}\left(\frac{\mathrm{i}g^2\mu^{6-d}}{(2\pi)^d}\right) \\ &\quad \times \int \mathrm{d}^dk \, \frac{1}{(k^2-m^2)[(p_1-k)^2-m^2][(p_2+k)^2-m^2]}. \end{aligned} \tag{218}$$

$$\begin{aligned} &\frac{1}{(k^2-m^2)[(p_1-k)^2-m^2][(p_2+k)^2-m^2]} \\ &\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \Big[(1-x-y)(k^2-m^2) + x((p_1-k)^2-m^2) + y((p_2+k)^2-m^2)\Big]^{-3} \end{aligned} \tag{219}$$

$$\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \left[k^2 - m^2 - 2(xp_1 - yp_2)\cdot k + xp_1^2 + yp_2^2\right]^{-3}. \tag{220}$$

Completing the square with $k' = k - xp_1 + yp_2$ and $P = p_1 + p_2$ gives

$$\begin{aligned} &2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \left[(k - xp_1 + yp_2)^2 - (xp_1 - yp_2)^2 + xp_1^2 + yp_2^2 - m^2\right]^{-3} \\ &\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \left[k'^2 - \Delta_b(x,y)\right]^{-3}, \end{aligned} \tag{221}$$

$$\Delta_b(x,y) = m^2 - p_1^2 x(1-x-y) - p_2^2 y(1-x-y) - P^2 xy. \tag{222}$$

After Wick rotation, the momentum integral is

$$\begin{aligned} &\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \int \mathrm{d}^dk_E \, \frac{2}{(k_E^2 + \Delta_b)^3} \\ &\quad = \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \frac{(2\pi)^d}{(4\pi)^{d/2}} \Gamma\left(3 - \frac{d}{2}\right) \Delta_b^{d/2-3}. \end{aligned} \tag{223}$$

Therefore

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right) \times \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3}. \tag{224}$$

The pole is especially transparent if we write $d = 6-\epsilon$. Since

$$\Gamma\left(3-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \qquad \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y = \frac{1}{2}, \tag{225}$$

the divergent part of graph (b) is

$$G_b^{\text{pole}} = S_{12}\frac{g^2}{64\pi^3}\frac{1}{\epsilon} = -S_{12}\frac{g^2}{64\pi^3(d-6)}. \tag{226}$$

Thus the operator counterterm that cancels this pole is local and proportional to the same two-leg insertion $\frac{1}{2}\phi^2$.

For graph (c), use the same P and set

$$D_{12P} = (p_1^2 - m^2)(p_2^2 - m^2)(P^2 - m^2). \tag{227}$$

Then

$$G_c = (-\mathrm{i}g\mu^{\frac{6-d}{2}})^2 \frac{-\mathrm{i}}{D_{12P}}\frac{1}{2!}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+P)^2-m^2} \tag{228}$$

$$= \frac{-\mathrm{i}g^2\mu^{6-d}}{D_{12P}}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k}{(2\pi)^d}\left[(k+xP)^2 - \Delta_c(x)\right]^{-2},$$

$$\Delta_c(x) = m^2 - P^2 x(1-x) \tag{229}$$

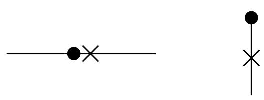
$$= \frac{\frac{1}{2}g^2\mu^{6-d}}{D_{12P}}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{1}{(k_E^2 + \Delta_c(x))^2} \tag{230}$$

$$= \frac{\frac{1}{2}g^2(\mu^2)^{3-d/2}}{D_{12P}}\frac{1}{(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\, \Delta_c(x)^{d/2-2} \tag{231}$$

$$= \frac{g^2}{D_{12P}}\frac{1}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{232}$$

$$G_c = \frac{-g\mu^{3-d/2}}{D_{12P}}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right) \times \int_0^1 \mathrm{d}x\, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{233}$$

G_b and G_c both diverge, so the bare composite operator ϕ^2 does not define finite Green functions by itself. For the OPE we need a local operator with the same quantum numbers and a finite insertion. In the MS scheme, the required counterterms are obtained by replacing each divergent short-distance loop by a local vertex:



For graph (b), the pole part is

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\int_0^{1-x}\mathrm{d}y\left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3} \tag{234}$$

$$= S_{12}\frac{g^2}{64\pi^3}\int_0^1 \mathrm{d}x\int_0^{1-x}\mathrm{d}y\left[\frac{2}{\epsilon}-\gamma_E-\ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right]+O(\epsilon) \tag{235}$$

$$= -S_{12}\frac{g^2}{64\pi^3(d-6)}+\text{finite}. \tag{236}$$

Therefore the corresponding MS counterterm insertion is

$$G_{b,\mathrm{ct}} = S_{12}\frac{g^2}{64\pi^3(d-6)}. \tag{237}$$

For graph (c), the pole part is

$$G_c = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}}, \tag{238}$$

where $\Delta_c(x) = m^2 - P^2x(1-x)$, $\qquad P = p_1 + p_2$, so

$$G_c = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\left(-\frac{2}{\epsilon}\right)\int_0^1 \mathrm{d}x\,\Delta_c(x)+\text{finite} \tag{239}$$

$$= \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right)+\text{finite}. \tag{240}$$

Thus

$$G_{c,\mathrm{ct}} = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\left\{\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right)\right\}. \tag{241}$$

This is exactly what is produced by inserting the local operator

$$\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2+\frac{1}{6}\Box\right)\phi. \tag{242}$$

Indeed, in momentum space $\Box$ acting on the local field gives $-P^2$, so $m^2+\Box/6$ becomes $m^2-P^2/6$. Throughout this subsection $d=6-\epsilon$, so $1/(d-6) = -1/\epsilon$; this is the source of the signs in the MS counterterms.

Thus, the renormalized value at $d=6$ is

$$R(G_b) = \frac{\mathrm{i}^2}{(p_1^2-m^2)(p_2^2-m^2)}\frac{g^2}{64\pi^3}\left[-\frac{\gamma_E}{2}-\int_0^1 \mathrm{d}x\int_0^{1-x}\mathrm{d}y\,\ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right], \tag{243}$$

$$R(G_c) = \frac{-g}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g}{128\pi^3}\left[(\gamma_E-1)\left(m^2-\frac{P^2}{6}\right)+\int_0^1 \mathrm{d}x\,\Delta_c(x)\ln\frac{\Delta_c(x)}{4\pi\mu^2}\right]. \tag{244}$$

The counterterms can now be read as local operator insertions. First, the two-leg insertion satisfies

$$\langle 0|\,\mathrm{T}()\{\tilde\phi(p_1)\tilde\phi(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\mathrm{conn}} = S_{12}. \tag{245}$$

Therefore the counterterm from graph (b) is equivalently

$$G_{b,\mathrm{ct}} = \frac{g^2}{64\pi^3(d-6)} \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\mathrm{conn}}\,. \tag{246}$$

Second, the one-leg operator insertion satisfies

$$\langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(0)\}\,|0\rangle_{O(g)} = -\frac{g}{D_{12P}}, \tag{247}$$

$$\langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2+\frac{1}{6}\Box\right)\phi(0)\}\,|0\rangle_{O(g)} = -\frac{g}{D_{12P}}\left(m^2-\frac{P^2}{6}\right). \tag{248}$$

Therefore the counterterm from graph (c) is equivalently

$$G_{c,\mathrm{ct}} = \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2+\frac{1}{6}\Box\right)\phi(0)\}\,|0\rangle_{O(g)}\,. \tag{249}$$

Graphs (d) and (e) have already been handled by the usual field and mass counterterms. The remaining new information is therefore the renormalization of the insertion itself:

$$\begin{aligned} &G_a + R(G_b) + R(G_c) + R(G_d) + R(G_e) \\ &= \left(1+\frac{g^2}{64\pi^3(d-6)}\right) \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle \\ &\quad + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2+\frac{1}{6}\Box\right)\phi(0)\}\,|0\rangle + \text{higher orders}. \end{aligned} \tag{250}$$

Thus the finite insertion is generated by the renormalized composite operator

$$\frac{1}{2}[\phi^2] \equiv \left(1+\frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2 + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2+\frac{1}{6}\Box\right)\phi + \text{higher orders} \tag{251}$$

$$= \left(1+\frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2 + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)}m^2\phi + \frac{g\mu^{d/2-3}}{6\cdot 64\pi^3(d-6)}\Box\phi + \text{higher orders}. \tag{252}$$

The brackets remind us that this is a renormalized local operator, not the bare product of two fields at the same point. To all orders, the same structure is written as

$$\frac{1}{2}[\phi^2] = Z_a\frac{1}{2}\phi^2 + \mu^{d/2-3}Z_b m^2\phi + \mu^{d/2-3}Z_c\Box\phi. \tag{253}$$

The constants Z_a, Z_b, Z_c contain only short-distance information; in minimal subtraction they are series in g and poles in $d-6$. The reason that only these operators appear is the same power-counting logic used before: counterterms must be local, and an operator can mix only with operators whose dimension is not larger than its own and whose quantum numbers match. In six-dimensional ϕ^3 theory, the allowed operators with the right dimension and quantum numbers are ϕ^2, $m^2\phi$, and $\Box\phi$. After ordinary field and mass renormalization has been performed, the remaining operator mixing can be summarized as

$$\begin{pmatrix} \frac{1}{2}[\phi^2] \\ \phi \\ \Box\phi \end{pmatrix} = \begin{pmatrix} Z_a & \mu^{d/2-3}Z_b m^2 & \mu^{d/2-3}Z_c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\phi^2 \\ \phi \\ \Box\phi \end{pmatrix}. \tag{254}$$